

YIRAN YANG

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EDUCATION

University of California, Davis

Expected 2030

Bachelor of Science in Mathematics & Computer Science

Davis, CA

- Relevant Coursework: Probability Theory, Game Theory, Statistical Inference, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Machine Learning, Causal Inference

SELECTED EXPERIENCE

Program in Mathematics for Young Scientists (PROMYS Italia)

Jul. 2025 – Aug. 2025

Selected Participant

Padua, Italy

- Selected as one of 17 participants from 200+ applicants (8.5% selection rate) on a €2,000 scholarship (~ 67% of €3,000).
- Solved 434 proof-based problems independently in Number Theory and Linear Algebra.
- Studied Linear Algebra and Representation Theory through university-level advanced seminars.

Non-Trivial Fellow

Jan. 2026 – Present

Selected Researcher

Remote

- Selected as one of 400 from 20,000+ global applicants for a 4-week program (~ 2% selection rate.); supported by Non-Trivial Team and Rethink Priority.
- Facilitated weekly one-on-one buddy calls to share research progress and refine decision frameworks.
- Building a causal inference framework to estimate marginal impact and optimize resource allocation.
- Leveraged knowledge in causal forests, dual machine learning, and linear programming for threshold detection and optimal allocation.

Comprehensive Oxford Mathematics and Physics Online School (COMPOS)

Jan. 2025 – Jun. 2026

Selected Participant

Remote

- Selected as one of 400 from 4,500+ global applicants (~ 8.9% selection rate) for a competitive year-long program.
- Led problem-solving sessions for mathematically gifted students worldwide.

Artificial Intelligence Youth Global Organization (AIYGO)

Apr. 2026 – Present

Senior Chair

Remote

- Managed 120+ members across 9 countries; designed quantitative reasoning materials on probability and risk concepts adopted by 90% of the Quantitative Finance committee members.
- Organized 6 monthly seminars on rational decision-making and risk analysis, each averaging 45+ attendees with guest speakers from leading universities.
- Mentored 8 peers in data-driven research, leading 3 projects focused on model validation, robustness testing, and decision framework design.

Leaf Courses (The Mathematics of Morality Course)

Jan. 2026 – Feb. 2026

Selected Participant

Remote

- Engaged in a rigorous 5-week intensive program, including seminars, and coursework via weekly discussion calls.
- Applied Bayesian inference, expected value theory, and Fermi estimation to build probabilistic cost-benefit models for comparing marginal returns across interventions.

RECENT AWARDS

Non-Trivial Fellowship Finalist (Top 200/20,000 globally, ~ 1%) | 2026

British Mathematical Olympiad Round 2 Qualifier & Oxford BMO Camp Invitation | 2025

Isaac Physics Senior Physics Challenge Emerald Award (Top 90/50,000+ globally, ~ 0.18%) | 2024

Millennium Fellowship Semi-Finalist | 2026

Susquehanna Discovery Event Participant | 2026

Optiver × COMPOS: Your Future with Math Participant | 2026

Leaf Courses Finalist (Top 50/1,500 globally, ~ 3.3%) | 2026

LANGUAGES AND SOFTWARE

Programming Skills (Proficient): L^AT_EX, Python

Frameworks and Tools (Proficient): Numpy, Pandas, PyTorch

Languages: Mandarin(Native), English(Working Proficiency)

Interests: Poker(No-Limit Texas Hold'em, Liar's Poker, etc.), Basketball, Ping Pong, Cooking, Travel